

One-operation completeness for polynomial rings: a characteristic-two dichotomy

— draft —

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Abstract

In Boolean logic a single operation, NAND, suffices to express every Boolean function. We ask the analogous question for polynomial rings: is there a single binary polynomial p such that every polynomial in two variables can be written as an iterated composition of p (starting from the variables and constants)? Over $\mathbb{Z}$ the answer is no: for every $p \in \mathbb{Z}[x, y]$, at least one of $x + y$, xy is inexpressible. More generally, a complete binary polynomial exists over a commutative ring R if and only if 2 is invertible in R : when it is, the single operation $p(x, y) = x^2 - y$ already generates everything; when it is not, completeness fails through a residue field of characteristic 2. The main work is the negative result over fields of characteristic 2, proved by an infinite descent whose decreasing quantity is, informally, the number of times a would-be counterexample factors through the squaring map. All results have been checked formally.

1 Introduction

1.1 A Sheffer stroke for $\mathbb{Z}[x, y]$?

Every Boolean function can be built from a single connective: NAND (the *Sheffer stroke*). One-element generating sets of this kind appear throughout algebra and logic, and it is natural to ask for an analogue one level up, with the two-element Boolean domain replaced by a ring of polynomials.

So fix the ring $\mathbb{Z}$ and consider polynomials in two variables. We are allowed to start from the raw materials

$$x, \quad y, \quad \text{and every constant } c \in \mathbb{Z},$$

and we are given *one* binary polynomial $p(x, y)$ as our only “connective”: whenever f and g have been built, we may build $p(f, g)$. Write $\text{Clo}(p)$ for the set of all polynomials obtainable this way (a precise definition is in Section 3; Section 2 explains how this relates to — and differs from — the standard notion of a *clone* in universal algebra, and no background from that subject is needed). Call p *complete* if $\text{Clo}(p) = \mathbb{Z}[x, y]$ — if the single operation p generates everything.

Could a complete p exist? Nothing obvious rules it out; a single polynomial has plenty of room to encode both an “additive” and a “multiplicative” mode, depending on what is

substituted into it. And since every polynomial is built from variables and constants by $+$ and $\times$, completeness is equivalent to just two memberships (Lemma 3.2):

$$p \text{ complete} \iff x + y \in \text{Clo}(p) \text{ and } xy \in \text{Clo}(p).$$

Our first result is that no single polynomial achieves even this.

Theorem 1.1. *For every $p \in \mathbb{Z}[x, y]$, at least one of $x + y$, xy lies outside $\text{Clo}(p)$. In particular, no binary polynomial over $\mathbb{Z}$ is complete.*

1.2 Why characteristic two

Where should such a theorem come from? The first observation is that completeness survives reduction of coefficients: a ring homomorphism $R \rightarrow S$ carries any derivation of $x + y$ over R to one over S . Reducing mod 2, Theorem 1.1 therefore follows from the corresponding statement over the two-element field $\mathbb{F}_2 = \mathbb{Z}/2$:

Theorem 1.2. *For every $q \in \mathbb{F}_2[x, y]$,*

$$x + y \notin \text{Clo}(q) \quad \text{or} \quad xy \notin \text{Clo}(q).$$

Two remarks on the statement. First, polynomials here are *formal* objects, not functions: over $\mathbb{F}_2$ infinitely many polynomials induce the same function on $\mathbb{F}_2^2$, and Theorem 1.2 concerns the polynomials themselves, so it does not follow from (and does not contradict) the classical theory of functionally complete operations on finite sets. Second — and this matters for the reduction — there is *no hypothesis on q whatsoever*. Reduction mod 2 can collapse degree ($2x^3 + xy \mapsto xy$), so to deduce Theorem 1.1 one needs the $\mathbb{F}_2$ -statement for completely arbitrary q , including the low-degree and degenerate ones.

Passing to $\mathbb{F}_2$ is not merely a convenience; characteristic 2 is where the obstruction actually lives. In Section 7 we will see that over any commutative ring in which 2 is invertible, the modest operation

$$p(x, y) = x^2 - y$$

is complete — a dozen lines of explicit algebra, valid over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Z}[\frac{1}{2}]$, and $\mathbb{F}_p$ for every odd prime p . (Over $\mathbb{F}_3$, for example, iterating $x^2 - y$ from the variables and constants produces *every* polynomial in $\mathbb{F}_3[x, y]$.) The same derivation, run over $\mathbb{Z}$, fails exactly at the steps that divide by 2; Theorem 1.1 says this failure is no accident of one derivation. Summarizing both directions:

Theorem 1.3 (the dichotomy). *Let R be a commutative ring. A complete binary polynomial over R exists if and only if 2 is a unit of R .*

The “only if” direction stacks three reductions on top of the $\mathbb{F}_2$ -argument. First, the descent behind Theorem 1.2 generalizes to every *perfect* field of characteristic 2 (one in which every element has a square root — all finite fields $\mathbb{F}_{2^k}$ and all algebraically closed fields, for instance), at the cost of a single twist explained in Remark 6.5; the disjunctive conclusion of Theorem 1.2 holds verbatim there. Second, an arbitrary field of characteristic 2 embeds into its perfection, and non-membership in the clone descends along coefficient embeddings, so in fact *no field of characteristic 2* admits a complete operation. Third, if 2 is not a unit of R , it lies in a maximal ideal; completeness pushes forward along the quotient map to the residue field, which has characteristic 2.

1.3 The shape of the proof, in words

The substantial result is Theorem 1.2, and the rest of this introduction previews its proof at leisure; the new idea is the descent in step 3, and we will take it slowly again when it appears for real in Section 6.

Step 1: a derivative splits the problem. Characteristic 2 kills second derivatives in each single variable ($\partial_x^2 = \partial_y^2 = 0$) but not the mixed one, so the operator $D := \partial_x \partial_y$ carries unusual weight. We split according to Dq .

If $Dq = 0$ — that is, no monomial of q has both exponents odd — then a pleasant chain-rule identity shows the property “ $Dq = 0$ ” is preserved by every substitution into q . The variables and constants have it; xy does not ($D(xy) = 1$). So $xy \notin \text{Clo}(q)$, and we are done with this case: a conserved quantity, nothing more.

If $Dq \neq 0$ we will show $x + y \notin \text{Clo}(q)$, and all the work lives here.

Step 2: from clones to curves. To exclude $x + y$ we use an invariant of a softer kind: a rigidity property of substitution. We will show that, when $Dq \neq 0$, a substitution $q(\alpha, \beta)$ can only land on a nonconstant polynomial in $x + y$ if α and β are themselves polynomials in $x + y$. A short induction over the clone (Proposition 5.1) shows this rigidity implies $x + y \notin \text{Clo}(q)$.

The rigidity property, in turn, is a statement about the *level sets* of q . Restricting everything to a generic line of the pencil $\{x + y = \text{const}\}$ converts it into the following concrete geometric assertion: there is no pair of one-variable polynomials $\alpha(t), \beta(t)$, not both constant, with

$$q(\alpha(t), \beta(t)) = c$$

identically in t , where c is a “generic” constant — precisely, a constant transcendental over $\mathbb{F}_2$, living in a large algebraically closed field K of characteristic 2. Such a pair (α, β) would be a polynomial-parametrized curve lying inside a single generic level set of q ; we call it a *witness*. The task is to show witnesses do not exist when $Dq \neq 0$.

Step 3: the descent. Here is the idea in one paragraph. Differentiate the witness equation $q(\alpha, \beta) = c$ with respect to t . In characteristic 2, derivatives interact with squares in extreme ways (squares have derivative zero; conversely, over a perfect field, anything with derivative zero *is* a square), and the derivative calculus along the witness forces a great deal of square structure. The endgame is a fork. Either some polynomial relation with $\mathbb{F}_2$ -coefficients holds along the witness — and any such relation is fatal, because it pins the witness to algebraic data, contradicting the genericity of the level c — or no such relation fires, in which case the square structure consumes the witness itself: both α and β turn out to be squares of polynomials of half the degree, $\alpha = \alpha_1^2$, $\beta = \beta_1^2$, and because the coefficients of q lie in $\mathbb{F}_2$ (where squaring fixes everything),

$$q(\alpha_1, \beta_1)^2 = q(\alpha_1^2, \beta_1^2) = c,$$

so (α_1, β_1) is again a witness, for the *same* q , at the level $\sqrt{c}$, with *half the degree*. Repeat. The degree cannot halve forever, so eventually the fatal branch fires.

The induction, note, is not on q at all: q never changes. It is on the witness, and the decreasing quantity is the number of times the witness can still factor through the Frobenius (squaring) map — what one might call its *Frobenius depth*. We found this inversion — inducting on the counterexample’s Frobenius depth rather than on any complexity measure of q

— to be the key that unlocked the problem, after many attempts at induction on degree-like measures of q failed. Degree of q is a poor induction variable here because substitution composes degrees in uncontrollable ways (note e.g. that q may reproduce low-degree polynomials from high-degree inputs); Frobenius depth of the witness is immune to this.

Because the argument has an unusual density of characteristic-2 bookkeeping, we have verified all results formally; a short note on this appears in Section 8, and the text otherwise presents proofs as ordinary mathematics.

2 Terminology, and relation to classical completeness theory

2.1 On the word “clone”

In universal algebra, a *clone* on a set A is a family of finitary operations $A^n \rightarrow A$ containing all coordinate projections and closed under composition; the classical theory — Post’s lattice and its descendants — studies such families of *functions* [6, 12]. Our $\text{Clo}(p)$ is not a clone in that sense, and we should say precisely what it is. Its elements are formal polynomials, not maps; the standard home for these is the *abstract clone* (equivalently, the Lawvere theory [3]) of the variety of commutative R -algebras, whose n -ary component is exactly the polynomial ring $R[x_1, \dots, x_n]$, with substitution as composition. Inside that abstract clone, $\text{Clo}(p)$ is the binary component of the smallest subclone containing p , the constants, and the projections. (Restricting to the binary component loses nothing: any term built from p , constants, and two variables is itself a bivariate polynomial, whatever arities its subterms pass through.) A reader who prefers to avoid clones altogether can take the low-tech view of Definition 3.1 at face value: $R[x, y]$ carries the single binary operation $(f, g) \mapsto p(f, g)$, and $\text{Clo}(p)$ is the subalgebra of this magma generated by $\{x, y\} \cup R$.

Two further cautions about nearby terminology. First, we admit all constants as generators; in the function-clone world this corresponds to *completeness with constants*. (The constant-free variant is also natural; we do not pursue it here.) Second, the phrase *polynomial completeness* has an established and quite different meaning in universal algebra — that every congruence-compatible function on an algebra is induced by a polynomial, as in the monograph [2] — and our usage of “complete” should not be confused with it.

2.2 Functional completeness, and why this question is different

For operations-as-functions on a finite set, one-element completeness is classical territory. Post’s classification of the Boolean clones locates the Sheffer stroke [10, 6]; Webb exhibited a single complete operation on every finite domain [13]; and there are effective completeness criteria, from Słupecki through Rousseau to Rosenberg’s classification of the maximal function clones [11, 9, 8].

How does the present question relate? Over an infinite integral domain a polynomial is determined by the function it induces, so over $\mathbb{Z}$ or $\mathbb{Q}$ our statements can equally be read in the function-clone world (on the infinite set $\mathbb{Z}$ or $\mathbb{Q}$, where the classical finite-domain theory is silent). Over finite fields, however, the formal question is strictly finer than the functional one, and the two can give *opposite verdicts on the same operation*:

Example 2.1. NAND is a polynomial: $n(x, y) := 1 + xy \in \mathbb{F}_2[x, y]$ induces the Sheffer stroke on $\{0, 1\}$, so the *function* clone it generates contains every Boolean function — functional completeness, no constants needed. But $D(1 + xy) = 1 \neq 0$, so the proof of Theorem 1.2 shows that the *polynomial* $x + y$ is not a composition of n : every NAND-formula whose induced function is XOR yields a polynomial that agrees with $x + y$ only as a function on $\mathbb{F}_2^2$, never on the nose (the difference is a nonzero polynomial vanishing identically on $\mathbb{F}_2^2$, e.g. a multiple of $x^2 + x$). Thus on $\mathbb{F}_2$, functional completeness is abundant while formal completeness is impossible: Theorem 1.2 is invisible to function-clone theory. In the other direction, over $\mathbb{F}_3$ our Theorem 7.1 produces every *polynomial*, which is strictly more than producing every function.

2.3 Composition structures on polynomial rings

The composition structure of polynomial rings has its own literature — Menger’s tri-operational algebras, Adler’s composition rings, and near-ring theory for univariate composition [4, 1, 5], as well as Ritt’s theory of decompositions of a single univariate polynomial [7]. These study the composition structure of the full polynomial ring rather than generation from a single binary operation, and we have not found the present question addressed there. We would be grateful for pointers to antecedents; the question is natural enough that we hesitate to claim novelty for anything but the answers.

3 Definitions and first reductions

Definition 3.1. Let R be a commutative ring and $p \in R[x, y]$. The *substitution clone* $\text{Clo}(p) \subseteq R[x, y]$ is the smallest set of polynomials such that

1. $x \in \text{Clo}(p)$, $y \in \text{Clo}(p)$, and $c \in \text{Clo}(p)$ for every constant $c \in R$;
2. if $f, g \in \text{Clo}(p)$ then $p(f, g) \in \text{Clo}(p)$,

where $p(f, g)$ denotes substitution of f for x and g for y in p . We call p *complete* over R if $\text{Clo}(p) = R[x, y]$.

Thus $\text{Clo}(p)$ is the set of polynomials expressible by a formula in the variables and constants whose only operation symbol is p .

Lemma 3.2. $\text{Clo}(p) = R[x, y]$ if and only if $x + y \in \text{Clo}(p)$ and $xy \in \text{Clo}(p)$.

Proof. One direction is trivial. For the other, first note that substituting members of the clone into a *member* of the clone again lands in the clone (induction on the formula). So if $x + y$ and xy are members, the clone is closed under the derived binary operations $(f, g) \mapsto f + g$ and $(f, g) \mapsto fg$; since it contains the variables and constants, it contains every polynomial. $\square$

Lemma 3.3 (transfer). *Let $\varphi : R \rightarrow S$ be a ring homomorphism and $\pi : R[x, y] \rightarrow S[x, y]$ the induced map on coefficients. Then $\pi(\text{Clo}(p)) \subseteq \text{Clo}(\pi p)$ for every $p \in R[x, y]$.*

Proof. π fixes the variables, sends constants to constants, and commutes with substitution; induct over the formula. $\square$

In particular, if both $x + y, xy \in \text{Clo}(p)$ over $\mathbb{Z}$, then both lie in $\text{Clo}(\pi p)$ over $\mathbb{F}_2$. So Theorem 1.2 (for *arbitrary* $q = \pi p$, degenerate cases included) implies Theorem 1.1, and likewise its analogue over any ring with a homomorphism to $\mathbb{F}_2$ ($\mathbb{Z}/2^k$, $\mathbb{F}_2$ -algebras, ...). The rest of the paper, until Section 7, takes place over $\mathbb{F}_2$.

We will constantly use three elementary facts about the formal partial derivatives ∂_x, ∂_y on $\mathbb{F}_2[x, y]$, all special to characteristic 2:

1. *Squares are derivative-free*: $\partial(f^2) = 2f \partial f = 0$.
2. *Repeated differentiation in one variable dies*: $\partial_x^2 = \partial_y^2 = 0$ (a monomial surviving ∂_x has had its odd x -exponent made even), while the mixed operator $D := \partial_x \partial_y$ does not.
3. *Frobenius*: squaring is a ring homomorphism, $(f+g)^2 = f^2 + g^2$; and a polynomial whose exponents are all even is a square (halve the exponents and use that every coefficient, being 0 or 1, is its own square root).

4 The derivative dichotomy over $\mathbb{F}_2$

Fix $q \in \mathbb{F}_2[x, y]$.

4.1 Parity decomposition

Group the monomials $x^i y^j$ of q into four classes by the parities of (i, j) . Each class is an even-exponent polynomial times one of $1, x, y, xy$, and even-exponent polynomials are squares (fact 3 above). Hence q can be written, uniquely, as

$$q = A_0^2 + xA_1^2 + yA_2^2 + xyA_3^2, \quad A_i \in \mathbb{F}_2[x, y]. \quad (1)$$

Differentiating (1) and discarding the squares (fact 1), only the prefixes x, y, xy contribute:

$$\partial_x q = A_1^2 + yA_3^2, \quad \partial_y q = A_2^2 + xA_3^2, \quad Dq = A_3^2. \quad (2)$$

So Dq is always a perfect square, and $Dq = 0$ exactly when q has no monomial with both exponents odd — equivalently, when $q \in \mathbb{F}_2[x^2, y] + \mathbb{F}_2[x, y^2]$, i.e. when q partially factors through the squaring map. The polynomial A_3 measures the failure of this degeneration, and the whole proof is organized around it.

4.2 The degenerate case $Dq = 0$: a conserved quantity

Lemma 4.1 (chain rule for D). *In characteristic 2, for all $q, \alpha, \beta \in \mathbb{F}_2[x, y]$:*

$$D(q(\alpha, \beta)) = (Dq)(\alpha, \beta) (\partial_x \alpha \partial_y \beta + \partial_y \alpha \partial_x \beta) + (\partial_x q)(\alpha, \beta) D\alpha + (\partial_y q)(\alpha, \beta) D\beta.$$

Proof sketch. Expand $\partial_x \partial_y (q(\alpha, \beta))$ by the ordinary chain and product rules. The terms that would involve $\partial_x^2 q$ or $\partial_y^2 q$ vanish by fact 2, and the survivors assemble as displayed. (In characteristic 0 the same computation produces additional second-derivative terms; their absence here is the point.) $\square$

Proposition 4.2. *If $Dq = 0$ then $xy \notin \text{Clo}(q)$.*

Proof. The set $\{g : Dg = 0\}$ contains the variables and constants, and by Lemma 4.1 (with the first term dead) it is closed under substitution into q . Hence it contains all of $\text{Clo}(q)$. But $D(xy) = 1$. $\square$

From now on suppose $Dq \neq 0$, i.e. $A_3 \neq 0$; we must show $x + y \notin \text{Clo}(q)$.

5 A rigidity property of substitution

Write $\mathbb{F}_2[x+y]$ for the subring of polynomials in $x + y$ (the “diagonal” polynomials). The proof of Theorem 1.2 in the case $Dq \neq 0$ runs through the following property of q :

($\dagger$) for all $\alpha, \beta \in \mathbb{F}_2[x, y]$: whenever $q(\alpha, \beta)$ is a *nonconstant* element of $\mathbb{F}_2[x+y]$, both $\alpha, \beta \in \mathbb{F}_2[x+y]$.

In words: the only way a substitution into q can land on a nonconstant diagonal polynomial is for both inputs to be diagonal already. The property is engineered to feed the following induction.

Proposition 5.1 (master reduction). *If q satisfies ($\dagger$) then $x + y \notin \text{Clo}(q)$.*

Proof. We claim every $g \in \text{Clo}(q)$ satisfies

$$P(g) : \quad g \notin \mathbb{F}_2[x+y] \quad \text{or} \quad g \text{ is constant.}$$

The variables satisfy P : the substitution $x \leftrightarrow y$ fixes every element of $\mathbb{F}_2[x+y]$ but moves x , so $x \notin \mathbb{F}_2[x+y]$, likewise y . Constants satisfy P . Inductive step: let $g = q(\alpha, \beta)$ with $P(\alpha), P(\beta)$, and suppose $P(g)$ fails: $g \in \mathbb{F}_2[x+y]$, nonconstant. Property ($\dagger$) forces $\alpha, \beta \in \mathbb{F}_2[x+y]$; then $P(\alpha), P(\beta)$ force α, β constant; but then g is constant, a contradiction. So P holds on the clone, and $x + y$ — which is in $\mathbb{F}_2[x+y]$ and nonconstant — is not in it. $\square$

So Theorem 1.2 reduces to:

Theorem 5.2. *If $Dq \neq 0$ then q satisfies ($\dagger$).*

5.1 From polynomials to curves

Property ($\dagger$) becomes a geometric statement after restriction to a generic member of the pencil of lines $\{x + y = \text{const}\}$. Fix once and for all an algebraically closed field $K \supseteq \mathbb{F}_2$ of characteristic 2 containing an element σ transcendental over $\mathbb{F}_2$; concretely, $K = \overline{\mathbb{F}_2}(\sigma)$, an algebraic closure of the field of rational functions in one indeterminate. We say $c \in K$ is *transcendental* if it satisfies no nonzero polynomial relation over $\mathbb{F}_2$ — the algebraic stand-in for “ c is a generic value”.

Definition 5.3. A *witness* is a pair $\alpha, \beta \in K[t]$, not both constant, such that $q(\alpha, \beta) = c$ is a constant, with c transcendental.

Geometrically: a polynomial-parametrized curve $t \mapsto (\alpha(t), \beta(t))$ in the plane over K , lying entirely inside the single level set $\{q = c\}$ of a generic level c .

Proposition 5.4 (bridge). *If no witness exists, then q satisfies $(\dagger)$.*

Proof. Let $\alpha, \beta \in \mathbb{F}_2[x, y]$ with $q(\alpha, \beta) = f(x + y)$, $f \in \mathbb{F}_2[u]$ nonconstant. Substitute the parametrized generic line $(x, y) = (t, t + \sigma)$, under which $x + y \mapsto \sigma$. The left side becomes $q(\alpha(t, t + \sigma), \beta(t, t + \sigma))$ and the right side the *constant* $f(\sigma)$, which is transcendental (a nonzero $\mathbb{F}_2$ -relation on $f(\sigma)$ composes with f to a nonzero relation on σ). Since no witness exists, both $\alpha(t, t + \sigma)$ and $\beta(t, t + \sigma)$ are constant in t . It remains to see that a polynomial $g \in \mathbb{F}_2[x, y]$ that is constant along the generic line lies in $\mathbb{F}_2[x + y]$: expand $g(x, x + u) = \sum_i g_i(u) x^i$ with $g_i \in \mathbb{F}_2[u]$; constancy in t of $g(t, t + \sigma)$ says $g_i(\sigma) = 0$ for all $i \geq 1$, hence $g_i = 0$ (σ transcendental), so $g(x, x + u) = g_0(u)$, i.e. $g = g_0(x + y)$ after substituting $u = y - x$ (which in characteristic 2 is $y + x$). $\square$

Theorem 5.2 has thus become a concrete statement we can attack with one-variable calculus:

Theorem 5.5 (no witnesses). *If $Dq \neq 0$, no witness exists.*

6 The Frobenius descent

This section proves Theorem 5.5. We first develop the mechanism in the simplest nontrivial situation, then do the general case.

Throughout, (α, β) is a putative witness: $\alpha, \beta \in K[t]$ not both constant, $q(\alpha, \beta) = c$, c transcendental. A prime $'$ denotes d/dt . Degrees are degrees in t , with the convention that constants — including the zero polynomial — have degree 0; under this convention $\deg(\alpha_1^2) = 2 \deg \alpha_1$ holds without exception, which keeps the bookkeeping of the descent honest. Recall the two calculus facts, now over $K[t]$: squares have zero derivative; and conversely, since K is algebraically closed (hence *perfect*: every element has a square root), a polynomial in $K[t]$ with zero derivative is a square — it is a polynomial in t^2 , and its coefficients have square roots.

6.1 Warm-up: the case $A_3 = 1$

Suppose first the parity decomposition (1) has $A_3 = 1$:

$$q = A_0^2 + xA_1^2 + yA_2^2 + xy, \quad \partial_x q = A_1^2 + y, \quad \partial_y q = A_2^2 + x.$$

(The smallest example is $q = xy$ itself; the case already contains all polynomials whose unique both-exponents-odd monomial is xy .)

Differentiate the witness equation $q(\alpha, \beta) = c$ in t . The chain rule gives

$$A\alpha' + B\beta' = 0, \quad A := (\partial_x q)(\alpha, \beta), \quad B := (\partial_y q)(\alpha, \beta). \quad (3)$$

Next differentiate A and B themselves. Using $\partial_x(\partial_x q) = 0$ and $\partial_y(\partial_x q) = Dq = 1$ (and symmetrically):

$$A' = \beta', \quad B' = \alpha'. \quad (4)$$

Combining (3) and (4):

$$(AB)' = A'B + AB' = \beta'B + A\alpha' = 0,$$

so AB is a *square* in $K[t]$. This is the engine: along a witness, the product of the two partial derivatives is forced to be a perfect square.

Now the fork. (For this warm-up we take for granted that neither A nor B vanishes identically; this degeneracy, like several others in the general case, is dispatched by the kill lemma below — Lemma 6.1 — and we do not belabor it here.) Suppose first A and B have a common root $\theta \in K$. Then the point $P := (\alpha(\theta), \beta(\theta))$ satisfies $\partial_x q(P) = \partial_y q(P) = 0$: it is a *critical point* of q . In the warm-up case the critical equations read $A_1^2(P) = \beta(\theta)$ and $A_2^2(P) = \alpha(\theta)$ — two polynomial equations over $\mathbb{F}_2$ pinning P down; one shows (this is easy here, and is Lemmas 6.3–6.2 in general) that P must have coordinates *algebraic* over $\mathbb{F}_2$. But then $c = q(P)$ is algebraic — contradicting transcendence of the level. Generic level sets pass through no critical points; that is what genericity is for.

Otherwise A and B are coprime in $K[t]$. A product of two coprime polynomials is a square only if each factor is (up to unit, and units of $K[t]$ are constants of the algebraically closed K , hence squares). So A and B are squares; squares have zero derivative; and (4) turns this into

$$\alpha' = \beta' = 0.$$

By perfectness, $\alpha = \alpha_1^2$ and $\beta = \beta_1^2$ for some $\alpha_1, \beta_1 \in K[t]$ of *half* the degree. And now the crucial use of $\mathbb{F}_2$ -coefficients: squaring fixes 0 and 1, so squaring a polynomial commutes with substituting into q ,

$$q(\alpha_1, \beta_1)^2 = q(\alpha_1^2, \beta_1^2) = q(\alpha, \beta) = c.$$

Writing $r := \sqrt{c} \in K$ (transcendental, since r algebraic would make $c = r^2$ algebraic), we conclude $q(\alpha_1, \beta_1) = r$: *the pair (α_1, β_1) is again a witness, for the same q , at half the total degree.*

Iterating, the total degree $\deg \alpha + \deg \beta$ halves at every round in the second branch, and cannot do so forever; the first branch must eventually fire, giving the contradiction. This proves Theorem 5.5 in the warm-up case. The induction variable, to repeat the point from the introduction, is a property of the *witness* — how many more times it can factor through squaring, its Frobenius depth — while q stands still.

6.2 What changes in general

For general q with $Dq = A_3^2 \neq 0$, two new things happen.

First, the engine identities (4) acquire the factor $\Delta := (Dq)(\alpha, \beta) = A_3(\alpha, \beta)^2$:

$$A' = \Delta \beta', \quad B' = \Delta \alpha',$$

which is harmless *provided* $\Delta \neq 0$ along the witness — a new way the argument can degenerate.

Second and more seriously, the fork's first branch needs critical points of q to be algebraic, and this can simply be false: the critical equations $\partial_x q = \partial_y q = 0$ may have *curves* of solutions. Indeed if $w := \gcd(A_1, A_2, A_3)$ is nonconstant then by (2) the whole curve $\{w = 0\}$ consists of critical points, and a witness might conceivably ride along near it. The remedy is to peel w off before running the engine. Write $A_i = wB_i$, so that the B_i have no common factor, and

$$q = A_0^2 + w^2 h, \quad h := xB_1^2 + yB_2^2 + xyB_3^2, \quad (5)$$

with $B_3 \neq 0$. We will run the warm-up argument *on h instead of q* : the function $H := h(\alpha, \beta)$ is no longer constant along the witness, but — and this is the small miracle that makes the peel legal — its derivative still vanishes. From (5) and $q(\alpha, \beta) = c = r^2$:

$$W^2 H = S^2, \quad W := w(\alpha, \beta), \quad S := r + A_0(\alpha, \beta), \quad (6)$$

using $(u + v)^2 = u^2 + v^2$. If $W \neq 0$, differentiating (6) (both W^2 and S^2 are squares) gives $W^2 H' = 0$, so $H' = 0$: *the moving level of h is Frobenius-flat*. The engine only ever used the vanishing of the level's derivative, so it runs on h unchanged.

What the peel buys is the algebraicity of critical points: for h , unlike q , the critical locus contains no curve. That is the keystone lemma below, and its proof is a pure characteristic-2 pleasantry. Finally, the various nondegeneracy conditions accumulated along the way ($W \neq 0$, $\Delta \neq 0$, $A \neq 0$, $B \neq 0$) are all of the same shape — some fixed nonzero polynomial with $\mathbb{F}_2$ -coefficients vanishing identically along the witness — and a single lemma disposes of them all: no such relation can hold, because it would tie the generic level to algebraic data.

We now state the three lemmas and assemble the proof.

6.3 Three lemmas

Lemma 6.1 (kill lemma). *Let (α, β) be a witness with level c , and $f \in \mathbb{F}_2[x, y]$, $f \neq 0$. Then $f(\alpha, \beta) \neq 0$ in $K[t]$.*

Proof sketch. Suppose $f(\alpha, \beta) = 0$; after swapping variables if needed, assume α nonconstant. Adjoin a fresh variable z standing for the level, and consider f and $g := q + z$ as polynomials in y over the ring $\mathbb{F}_2[x, z]$. Since g is monic and linear in z , it is irreducible and cannot divide f (which is z -free), so f and g are coprime; a Bézout identity over the fraction field, with denominators cleared, produces

$$fU + (q + z)V = E(x, z) \neq 0 \quad \text{in } \mathbb{F}_2[x, z][y].$$

Now substitute $y \mapsto \beta$, $x \mapsto \alpha$, $z \mapsto c$: the first summand dies by hypothesis, the second because $q(\alpha, \beta) + c = c + c = 0$ in characteristic 2. So $E(\alpha, c) = 0$. Expanding $E = \sum_i \rho_i(z) x^i$ and using that $1, \alpha, \alpha^2, \dots$ have strictly increasing degrees in t (here nonconstancy of α enters), each coefficient $\rho_i(c)$ must vanish; some $\rho_i \neq 0$ then exhibits c as algebraic over $\mathbb{F}_2$ — contradiction. $\square$

Lemma 6.2 (algebraic points). *If $u, v \in \mathbb{F}_2[x, y]$ have no common nonunit factor and $(a, b) \in K^2$ satisfies $u(a, b) = v(a, b) = 0$, then a, b are algebraic over $\mathbb{F}_2$.*

Proof sketch. Viewing u, v in $\mathbb{F}_2(x)[y]$: a common divisor there would descend (Gauss's lemma) to a common factor in $\mathbb{F}_2[x, y]$, so they are coprime, and a cleared Bézout identity gives $su + tv = r(x) \neq 0$; evaluate at (a, b) to get $r(a) = 0$. Symmetrically for b . $\square$

Lemma 6.3 (keystone: no curve of critical points after the peel). *Let $B_1, B_2, B_3 \in \mathbb{F}_2[x, y]$ have no common nonunit factor, with $B_3 \neq 0$, and set $h_x := B_1^2 + yB_3^2$, $h_y := B_2^2 + xB_3^2$ (the partials of h). Then h_x and h_y have no common nonunit factor.*

Proof. Both are nonzero: applying ∂_y to a relation $h_x = 0$ would give $B_3^2 = 0$. Suppose a prime p divides both.

If $p \mid B_3$: subtracting the B_3^2 -terms, p divides B_1^2 and B_2^2 , hence (primality) all three B_i — excluded.

If $p \nmid B_3$: write $h_x = pu$ and $h_y = pv$. Differentiate $h_x = B_1^2 + yB_3^2$, remembering squares are derivative-free:

$$0 = \partial_x h_x = p_x u + p u_x \implies p \mid p_x u;$$

$$B_3^2 = \partial_y h_x = p_y u + p u_y \implies p \nmid u$$

(else $p \mid B_3^2$, so $p \mid B_3$);

$$B_3^2 = \partial_x h_y = p_x v + p v_x \implies p \nmid p_x$$

(same reason). So the prime p divides the product $p_x u$ but neither factor — contradiction. $\square$

Remark 6.4. Lemma 6.3 has geometric content: a curve of common zeros of h_x, h_y would have a function field in which both x and y become squares (solve the critical equations for y and x), making that function field *perfect*; but the function field of a curve in characteristic 2 is never perfect. The displayed proof is this argument in infinitesimal form — “ $p \nmid p_x$ ” is the derivation that survives — and entirely avoids the geometry.

6.4 Proof of Theorem 5.5

Among all witnesses, take one minimizing $n := \deg \alpha + \deg \beta$ (≥ 1 , as not both are constant). Decompose q as in (5); fix $r = \sqrt{c}$, transcendental.

By the kill lemma (Lemma 6.1) applied to each of the nonzero polynomials w, h_x, h_y, B_3 — nonzero by construction and by the argument opening Lemma 6.3 — none of

$$W := w(\alpha, \beta), \quad A := h_x(\alpha, \beta), \quad B := h_y(\alpha, \beta), \quad \Delta := B_3(\alpha, \beta)^2$$

vanishes in $K[t]$.

Peel. As in (6), $W^2 H = S^2$ with $H = h(\alpha, \beta)$, $S = r + A_0(\alpha, \beta)$, and differentiating, $H' = 0$.

Engine. The chain rule for h , plus $\partial_x h_x = 0$, $\partial_y h_x = \partial_x h_y = Dh = B_3^2$, $\partial_y h_y = 0$, gives

$$A\alpha' + B\beta' = H' = 0, \quad A' = \Delta\beta', \quad B' = \Delta\alpha',$$

whence $(AB)' = \Delta(A\alpha' + B\beta') = 0$ and AB is a square.

Fork, branch 1: A and B share a root θ . Then $P := (\alpha(\theta), \beta(\theta))$ is a common zero of h_x, h_y , which by Lemmas 6.3 and 6.2 has algebraic coordinates. Evaluate the peel identity (6) at θ :

$$S(\theta)^2 = w(P)^2 h(P)$$

is algebraic, hence so is $S(\theta)$, hence so is $r = S(\theta) + A_0(P)$. This contradicts transcendence of r .

Fork, branch 2: A, B coprime. Coprime factors of a square are squares, so $A' = B' = 0$; since $\Delta \neq 0$, the engine identities force $\alpha' = \beta' = 0$, so $\alpha = \alpha_1^2$, $\beta = \beta_1^2$, and as in the warm-up $q(\alpha_1, \beta_1) = r$. Then (α_1, β_1) is a witness with $\deg \alpha_1 + \deg \beta_1 = n/2 < n$, contradicting minimality. (If $n = 1$ this branch cannot occur at all, as halving is impossible; so the descent is well-founded.) $\square$

Remark 6.5 (from $\mathbb{F}_2$ to the dichotomy). The descent generalizes to any perfect coefficient field k of characteristic 2 with one modification. The coefficients of q are no longer fixed by squaring, so in the halving branch the identity $q(\alpha_1^2, \beta_1^2) = \tilde{q}(\alpha_1, \beta_1)^2$ holds not for q itself but for the *coefficient-twisted* polynomial $\tilde{q} = q^{(1/2)}$ (apply the inverse Frobenius to each coefficient — this is where perfectness enters). The halving branch therefore produces a half-degree witness for $q^{(1/2)}$ rather than for q ; since the induction is on the witness degree with q universally quantified inside, the twist is absorbed at no cost, and no periodicity argument is needed. For an arbitrary field k of characteristic 2, embed k into its perfection $k^{1/2^\infty}$: a derivation of $x + y$ (or xy) over k maps to one over the perfection, contradicting the perfect case. Finally, for a ring R in which 2 is not a unit, 2 lies in a maximal ideal $\mathfrak{m}$, and a complete operation over R would push forward along $R \rightarrow R/\mathfrak{m}$ to a complete operation over a field of characteristic 2. This proves Theorem 1.3.

Assembling: Theorem 5.5 $\Rightarrow$ (bridge, Proposition 5.4) property $(\dagger) \Rightarrow$ (master reduction, Proposition 5.1) $x + y \notin \text{Clo}(q)$ when $Dq \neq 0$; together with Proposition 4.2 for $Dq = 0$, this proves Theorem 1.2, and with Lemma 3.3, Theorem 1.1. $\square$

7 Aftermath: completeness when 2 is invertible

We now show characteristic 2 is the only obstruction, by exhibiting a complete operation everywhere else. This section is independent of Sections 4–6.

Theorem 7.1. *Let R be a commutative ring with $\frac{1}{2} \in R$. Then $q(x, y) := x^2 - y$ is complete: $\text{Clo}(x^2 - y) = R[x, y]$.*

Proof. By Lemma 3.2 it suffices to reach $x + y$ and xy . Everything flows from one identity:

$$q(\alpha, q(\alpha', \gamma)) = \alpha^2 - \alpha'^2 + \gamma. \quad (7)$$

Translations. With constants $\alpha = \frac{d+1}{2}$, $\alpha' = \frac{d-1}{2}$, identity (7) sends $\gamma \mapsto \gamma + d$, for any prescribed $d \in R$. (This is the only step that uses $\frac{1}{2}$.) In particular $x + d, y + d \in \text{Clo}(q)$ for all d .

Slanted lines. $q(c, y) = c^2 - y$, then $q(x, c^2 - y) = x^2 + y - c^2$, then

$$q(x + d, x^2 + y - c^2) = 2dx - y + (d^2 + c^2) \in \text{Clo}(q),$$

a line of slope $2d$ for every d , with constant adjustable by translations.

Linear engine. For any $\alpha \in \text{Clo}(q)$ and constant k , apply (7) with outer argument $\alpha + k$ and inner argument α : the increment is

$$\gamma \mapsto (\alpha + k)^2 - \alpha^2 + \gamma = \gamma + 2k\alpha + k^2.$$

Increments along slanted lines of two distinct slopes span all linear polynomials, so starting from $\gamma = 0$: every linear polynomial lies in $\text{Clo}(q)$, in particular $x + y$ and $x + \frac{y}{2}$.

Scaled squares. The same engine with $\alpha = y^2 + \ell$ (in the clone via $q(y, -\ell)$) adds $\lambda y^2 + \text{const}$ for any λ ; from $\gamma = x^2 = q(x, 0)$ we get $x^2 + \frac{1}{4}y^2 \in \text{Clo}(q)$.

Finish.

$$q\left(x + \frac{y}{2}, x^2 + \frac{1}{4}y^2\right) = \left(x + \frac{y}{2}\right)^2 - x^2 - \frac{1}{4}y^2 = xy. \quad \square$$

Remark 7.2. All constants used lie in $\mathbb{Z}[\frac{1}{2}]$, so the derivation runs verbatim over every ring in which 2 is invertible — in particular over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Z}[\frac{1}{2}]$, and $\mathbb{F}_p$ for odd p ; over $\mathbb{F}_3$, the single operation $x^2 - y$ generates the full polynomial ring $\mathbb{F}_3[x, y]$. Over $\mathbb{Z}$ the translation step requires realizing $d = c^2 - e^2$ with $c, e \in \mathbb{Z}$, which is possible only for $d \not\equiv 2 \pmod{4}$; the increments available are then too even to span, and the derivation jams — as, by Theorem 1.1, every derivation must.

Remark 7.3. Theorem 7.1 produces a complete operation of degree 2. We do not know the complete picture for higher degrees; see Section 9.

8 A note on formal verification

The proofs above, especially the descent, involve a large amount of routine characteristic-2 manipulation, and we wanted an independent check. All results stated here — Theorems 1.1, 1.2, 7.1, 1.3 and the supporting lemmas — have been verified in the Lean 4 proof assistant against the Mathlib library, with no unproved assumptions beyond Lean’s standard foundational axioms. The formal statements involve only the fifteen-line definition of Clo reproduced in Appendix A. Two of the proofs presented above are actually simplifications found during the verification process: the derivative proof of the keystone (Lemma 6.3), which replaced a longer argument via perfect function fields, and the Bézout-certificate form of the kill lemma. We will not say more about the formalization here; the development is available from the author.

9 Open questions

Question 9.1. Which degrees admit complete operations over $\mathbb{Q}$? Degree 2 does (Theorem 7.1). For degree 3 a natural candidate is $x^3 - y$: translations by arbitrary rationals follow from the classical fact that every rational number is a sum of three rational cubes (with explicit formulas), and a cubic analogue of the linear engine appears to reach xy ; we have not completed the verification. The answer is characteristic-sensitive: over $\mathbb{F}_3$ the map $u \mapsto u^3$ is additive (Frobenius), and one checks that the clone of $x^3 - y$ consists only of polynomials of the form *additive plus constant* — $\mathbb{F}_3$ -linear combinations of the monomials x^{3^i}, y^{3^j} , plus a constant (the constants enter because they are generators) — a class closed under substitution into $x^3 - y$ and excluding xy ; so $x^3 - y$ is certainly not complete there.

Question 9.2. Away from characteristic 2, *which* operations are complete? Completeness is not simply “nonlinear and depending on both variables”: over $\mathbb{Q}$,

- $xy + 1$ is not complete, although it is nonlinear, uses both variables, and its clone contains xy . The obstruction is a characteristic-zero instance of the rigidity property (†): if $\alpha\beta + 1 = F(x+y)$ is nonconstant then $\alpha\beta = h(x+y)$ splits over $\overline{\mathbb{Q}}$ into linear factors $(x+y - r_i)$, so by unique factorization every divisor — in particular α and β — lies in $\mathbb{Q}[x+y]$; the master reduction (whose proof is characteristic-free) then excludes $x + y$.
- $x^2 + y$ is not complete, although $x^2 - y$ is: every element of $\text{Clo}(x^2 + y)$ has the form $a + \sum_i s_i^2$ with $a \in \{x, y\} \cup \mathbb{Q}$ (the shape is preserved by $(f, g) \mapsto f^2 + g$), and neither

$x + y$ nor xy is of this form, since sums of squares of real polynomials take no negative values. A *positivity* obstruction from the real place, invisible over finite fields.

Together with the affine obstruction (linear p) and the monomial obstruction ($\text{Clo}(xy)$ consists of monomials), this suggests asking whether completeness is governed by a finite explicit list of “conserved classes” — proper subsets of $R[x, y]$ containing the generators and closed under substitution into p . We do not know; we also cannot presently decide simple boundary cases such as $x^2 - 2y$, where the engine increment $\gamma \mapsto 4\gamma + (\alpha^2 - 2\alpha'^2)$ rescales instead of translating and a 2-adic conserved class is conceivable. A positive structure theorem here would likely also bear on Question 9.4.

Question 9.3. The same questions can be posed for the field of rational functions: preliminary calculations suggest $1/x - y$ is complete over $\mathbb{Q}(x, y)$ in a suitable sense. Substitution of rational functions is a partial operation (denominators may vanish identically), so even stating this precisely requires a partial-clone formalism.

Question 9.4. Is membership in $\text{Clo}(p)$ decidable? Is completeness?

Half of the problem is easy: membership is semidecidable, since one can enumerate all formulas in p and expand each. It is the complementary half that we want to dwell on, because the difficulty is structural and shaped our whole approach. The obvious idea — search all formulas up to some size or degree bound, and conclude non-membership if the target never appears — proves nothing, no matter the bound, because substitution is not monotone in any evident complexity measure. The culprit is *cancellation*. Already for $q = x^2 + y$ over $\mathbb{F}_2$ one has, for *all* f, g ,

$$q(f, q(f, g)) = f^2 + (f^2 + g) = g :$$

the intermediate polynomial f , of arbitrarily large degree, is annihilated without trace. So every element of the clone admits derivations passing through intermediates of unbounded degree, and a small target may, for all a bounded search can tell, first become reachable only after a long excursion through enormous polynomials that cancel at the last step. The degree of what you are building says nothing about the degree of what you must build it from.

This is why every non-membership argument in this paper avoids search entirely and instead exhibits a *cancellation-immune invariant*: a property that is checked on the final polynomial alone and is preserved by every substitution into q , however the intermediate terms conspire — membership in $\ker D$ in Proposition 4.2, the invariant P in Proposition 5.1. A decision procedure would seem to require either a computable bound on derivation size, which cancellation conspires against, or a supply of such invariants complete enough to certify every non-membership; we possess neither, and we suspect the question is hard. As a concrete instance of how little we know: for q with $Dq = 0$, Theorem 1.2 settles $xy \notin \text{Clo}(q)$, but we know no algorithm deciding, for such q , whether $x + y \in \text{Clo}(q)$.

Draft note. The references below are reconstructed from memory and must be verified (and surely extended) before this goes anywhere; corrections welcome.

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A The formal statement layer

For the record, the complete definitional trust base of the formal verification (R an arbitrary commutative ring; $X\ 0, X\ 1$ the two variables of a bivariate polynomial ring; $\text{bind1}\ ! [a, b]$ p denotes the substitution $p(a, b)$):

```

inductive Clo (p : MvPolynomial (Fin 2) R) :
  MvPolynomial (Fin 2) R -> Prop
| atomX : Clo p (X 0)
| atomY : Clo p (X 1)
| const (c : R) : Clo p (C c)
| sub {a b : MvPolynomial (Fin 2) R} :
  Clo p a -> Clo p b -> Clo p (bind1 ! [a, b] p)

theorem F2_master (q : MvPolynomial (Fin 2) (ZMod 2)) :
```



```

~ Clo q (X 0 + X 1) \ / ~ Clo q (X 0 * X 1)

theorem int_master (p : MvPolynomial (Fin 2) Int) :
  ~ Clo p (X 0 + X 1) \ / ~ Clo p (X 0 * X 1)

theorem qOp_complete {R} [CommRing R] [Invertible (2 : R)] :
  forall g, Clo (X 0 ^ 2 - X 1 : MvPolynomial (Fin 2) R) g

theorem complete_iff_two_isUnit (R) [CommRing R] :
  (exists p : MvPolynomial (Fin 2) R, forall g, Clo p g)
  <-> IsUnit (2 : R)

```